

cl-freelock Documentation

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1 Tutorial

1.1 Getting Started: Installation

Once available on quicklisp, you will be able to simply run:

```
(ql:quickload :cl-freelock)
```

For now, clone the repository into your Quicklisp local-projects directory:

```
cd ~/quicklisp/local-projects/  
git clone https://github.com/ItsMeForLua/cl-freelock.git
```

Then, load the system in your REPL:

```
(ql:quickload :cl-freelock)
```

1.2 Basic Usage

In this section, we will go over the basic usage of producer-consumer with unbounded queue, bounded buffers, SPSC, error handling, and atomic operations.

FYI

1. `cl-freelock` allows the use of `fl` when calling its functions.

For example: `(fl:queue-push *work-queue* i)`

2. The API reference for `cl-freelock` is mostly written with SBCL in-mind. However, the differences between implementations regarding the code presented in this documentation will be quite minor.

1.2.1 MPMC Unbounded Queue (Michael-Scott)

The algorithm used for unbounded queues is based off of the Michael-Scott algorithm. We specifically use Compare-And-Swap (CAS) operations to link nodes and swing the head and tail pointers.

We identify three main reasons one would use this particular queue type:

1. When the workload is highly variable in predictability, such that you cannot guarantee a maximum capacity for your queue ahead of time.
2. You have a thread pool architecture with multiple producer threads that are throwing jobs onto the queue, and multiple worker threads are simultaneously pulling jobs off. i.e., An N-to-N architecture.
3. If your project requires immunity from deadlocks.

Producers

A producer thread works by calling `(queue-push *work-queue* item)`. Because the producer thread is lock-free, the procedure will never block a thread. An example of defining a producer:

```
(defun producer ()  
  (loop for i from 1 to 1000 do  
    (fl:queue-push *work-queue* i)  
    (sleep 0.001)))
```

n.b., It is very important you consider the reasoning for adding a sleep in this example. The sleep is just to simulate the time it may take to fetch the next item. It is not needed in the procedure within normal use-cases; however, if you simply ran the example in a fresh REPL, you would risk locking up your entire environment.^{1, 2}

1. This is called “CPU Starvation”, which is when a procedure consumes 100% of a systems CPU core.
2. Concurrent programming is a matter of managing the interleaving of different processes as they interact with a shared state – hence, if we ran a loop from i to 1000 without sleep, most modern CPUs would actually

Consumers

A consumer thread works by calling `(queue-pop *work-queue*)`. This procedure does not return lisp conditions; instead, `queue-pop` returns two specific values:

1. The `item` itself or `NIL` if empty.
2. A `success` boolean, `T` if an item was grabbed, `NIL` if the queue was empty.

n.b., If a queue happens to be empty, then the consumer thread should handle it gracefully by briefly sleeping before polling again. The sleep is a small number, which you set to prevent busy waiting.³

You must wrap the procedure in a `multiple-value-bind` to check if the pop was successful.

```
(defparameter *work-queue* (make-queue))
;; Producer thread
(defun producer ()
  (loop for i from 1 to 1000 do
    (queue-push *work-queue* i)
    (sleep 0.001)))

;; Consumer thread
(defun consumer ()
  (loop
    (multiple-value-bind (item found)
      (queue-pop *work-queue*)
      (if found
        (format t "Processing: ~A~%" item)
        (sleep 0.001))))))
```

1.3 MPMC Bounded Queue (Vyukov)

be so fast that the producer may finish pushing all 1000 items before a consumer thread has a chance to wakeup.

3. Busy waiting (or spinning) is when a consumer constantly asks an empty queue if it has any data in queue – consequently, this tanks a systems performance.

```

(defparameter *buffer* (make-bounded-queue 64)) ; Must be power of 2

;; Non-blocking operations
(when (bounded-queue-push *buffer* "data")
  (format t "Push succeeded~%"))

(multiple-value-bind (value success)
  (bounded-queue-pop *buffer*))
(when success
  (format t "Got: ~A~%" value)))

```

1.4 SPSC Bounded Queue

```

(defparameter *spsc* (make-spsc-queue 256)) ; Must be power of 2
;; Producer thread only
(when (spsc-push *spsc* "fast-data")
  (format t "Pushed successfully~%"))
;; Consumer thread only
(multiple-value-bind (value success)
  (spsc-pop *spsc*))
(when success
  (format t "Got: ~A~%" value)))

```

1.5 Error Handling

Operations return multiple values to indicate success/failure rather than signaling conditions:

```

;; Always check the second return value
(multiple-value-bind (value success)
  (queue-pop *queue*))
(if success
  (process value)
  (handle-empty-queue)))

```

2 API Reference

2.1 Atomic Operations

2.1.1 ATOMIC-REF

2.1.2 MAKE-ATOMIC-REF

2.1.3 ATOMIC-REF-VALUE

2.1.4 CAS

2.1.5 ATOMIC-INCF

2.1.6 MEMORY-BARRIER

2.2 UMPMC Unbounded Queue

2.2.1 QUEUE

Lock-free unbounded queue supporting multiple producers and consumers.

2.2.2 MAKE-QUEUE

2.2.3 QUEUE-PUSH

2.2.4 QUEUE-POP

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2.3 MPMC Bounded Queue

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2.4 SPSC Bounded Queue

2.4.1 SPSC-QUEUE

2.4.2 MAKE-SPSC-QUEUE

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2.4.4 SPSC-POP

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